PT4052 - PT7004 PRACTICAL:

MSK: Exercise prescription - post orthopaedics procedures

(Group A & B)

Date: Tue 12/11/2024 **Time: 14:00 - 15:30 & 15:30 – 17:00**

Learning Outcomes:

- Prescribe Exercises after orthopaedic procedures for the lower limb.
- Understand Precautions/Contraindications.
- Identify open and closed kinetic chain exercises.
- Develop clinical reasoning skills for early exercise prescription post orthopaedic procedures (for total hip replacement, total knee replacement and ankle fracture mobilisation).

Post-Hip Replacement:

- Demonstrate the following exercises in small groups, deep breathing, circulatory exercises, hip flexion, hip abduction, static quads, static gluts, mobilising.
- Identify if the exercise is an open or closed kinetic chain:
- Explain why the exercise is beneficial.

Timeline of subsequent THR Rehabilitation

Post-knee replacement:

- Movements not to do after surgery:
- Early Stages: Deep Breathing, Circulatory exercises, Static quads, Straight leg raise, Static hamstrings, Static gluts, Mobilising, Inner Range Quads, Full range Quads, Knee flexion in sitting.
- Timeline of Subsequent TKR Rehabilitation.

Post Ankle Replacement:

- Early Exercises: Foot pumps, ankle circles, heel raises, assisted plantarflexion, dorsiflexion, inversion, eversion,

Early exercises – These are designed to help improve movement and function. They should be started as soon as your plaster is removed and carried out three to four times a day.
Intermediate exercises – Only start these exercises if your consultant is happy for you to put full weight on your foot: Resisted dorsiflexion, resisted plantarflexion, internal rotation (/eversion), external rotation (/inversion), single leg balance.

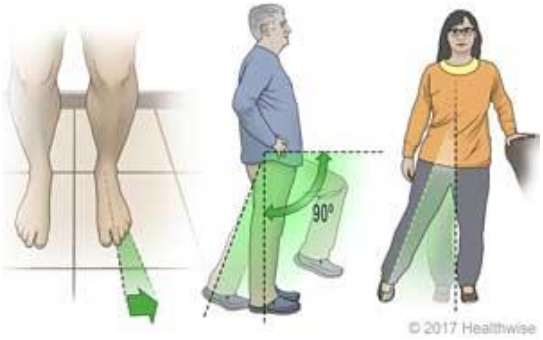
Exercise Prescription Post Orthopaedic Procedures

Learning Outcomes:
By the end of this session students will be able to: • Prescribe Exercises after orthopaedic procedures for the lower limb. • Understand Precautions/Contraindications. • Identify open and closed kinetic chain exercises. • Develop clinical reasoning skills for early exercise prescription post orthopaedic procedures (for total hip replacement, total knee replacement and ankle fracture mobilisation).

Post-hip replacement:

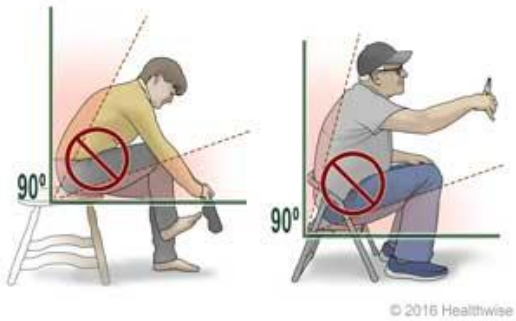
Video: https://www.youtube.com/watch?v=JyPp3spPKiE&t=463s

What are some precautions for self-care after hip replacement surgery (posterior)?



Some precautions for self-care after hip replacement surgery.
Keep your toes pointing forward or slightly out. Don't rotate your leg too far to the inside.
Do not bend your hip more than 90 degrees.
Keep your knees apart. Don't cross your legs.

Hip Replacement (Posterior) Precautions: Don't bend your hip too far



Hip Replacement Precautions:
Don't bend your hip too far
Don't lean forward while you sit down or stand up, and don't bend past 90 degrees (like the angle in a letter "L"). This means you can't try to pick up something off the floor or bend down to tie your shoes.
Don't lift your knee higher than your hip.
Don't sit on low chairs, beds, or toilets. You may want to use a raised toilet seat for a while.
Sit in chairs with arms.

Hip Replacement (Posterior) Precautions: Don't cross your legs



Hip Replacement Precautions: Don't cross your legs

Hip Practical –

- Demonstrate the following exercises in small groups
- Identify if the exercise is an open or closed kinetic chain:
- Explain why the exercise is beneficial.

Early stages:

Exercise	Open / Closed Kinetic Chain	Clinical Reasoning
Deep Breathing		
Circulatory exercises		
Hip flexion on the bed		
Hip abduction on the bed		
Static quads		
Static gluts		
Mobilising		

Dependent on patient presentation:

Exercise	Open / Closed Kinetic Chain	Clinical Reasoning
Full range quads		
Knee flexion in sitting		
Inner range quads		
Abduction in standing		
Hip flexion in standing		
Hip extension in standing		

Timeline of subsequent THR Rehabilitation

2-3 weeks post-op

Half squats Heel raises in standing

4-6 weeks post-op

Static bike Marching on the spot Half squats Hip abduction in standing Hip extension in standing External rotation in standing Walking exercises • Walking forwards • Walking backwards • Walking sideways Pool / Hydro • Hip extension • Knee towards chest • Hip abduction Advanced pool exercises: • Jumping • Crunch jumps • Running on the spot

6-8 weeks

Single leg balance Step ups Step downs Hip extension in prone External rotation against a wall Bridging Clam shell Hip abduction in side lying

6 months

You should now be back to full activities with the exception of high impact sports.

1 year

You should now be fully recovered and able to carry out all activities of daily living without problems.

Post-knee replacement:

Video: https://www.youtube.com/watch?v=_wwnpMuG380

Movements not to do after surgery

It's important to ease back into activities after surgery. But there are some movements you should **not** do for the first 3 months after surgery.

- **Don't** jar or twist your new knee suddenly. Make sure you don't bend it in an uncontrolled way.
- **Don't** cross your legs.
- **Don't** sleep with a pillow under your knee. It can cause a permanent bend in your knee or put pressure on blood vessels in your leg.

When you twist or turn:

- Always keep your toes and your upper body facing the same direction.
- Take small steps when turning, instead of twisting or pivoting on your leg.

Knee Practical –

Early stages:

Exercise	Open / Closed Kinetic Chain	Clinical Reasoning
Deep Breathing		
Circulatory exercises		
Static quads		
Straight leg raise		
Static hamstrings		
Static gluts		
Mobilising		

Dependent on patient presentation:

Exercise	Open / Closed Kinetic Chain	Clinical Reasoning
Inner range quads		
Full range quads		
Knee flexion in sitting		

Timeline of subsequent TKR Rehabilitation

Week 2+

Patella mobs Assisted knee bend in sitting Resisted exercises in sitting Sit in a chair, bend your knee as far as possible. Passive hyperextension Heel raises in standing Half squats Knee flexion in standing Step ups Step downs Heel digs Single leg balance Single leg heel raise

Week 4+

Static bicycle if available. Balancing with feet together Balancing one foot in front of the other Rolling a ball forward and back while seated Rolling ball in small circles while seated Squashing ball into the floor Squeeze your knees together Rolling ball while standing Hydro / Pool Marching on the spot Knee flexion in standing Half squats Water walking exercises • Walking forwards • Walking sideways • High knee walking: walk forward • Lunge walking • Toe walking

1 – 3 months

Increase repetitions or resistance of exercises (kg or resistance band)

3 – 6 months

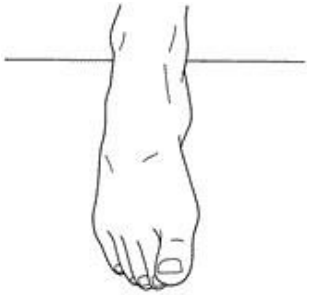
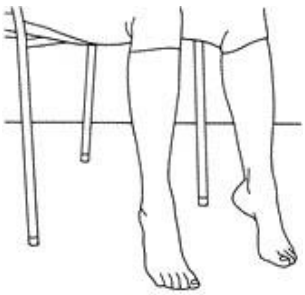
Able to return to low impact sports, but avoid high impact sports involving running, jumping or contact. May still have some joint line pain.

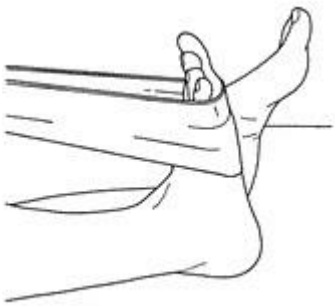
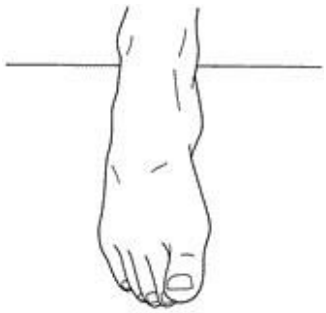
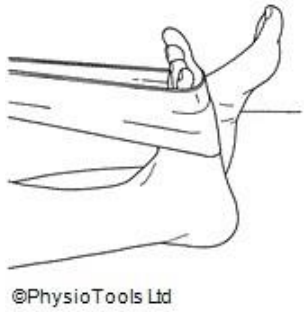
6 months – 1 year

Normal Activities. Joint line pain can persist for up to two years after surgery.

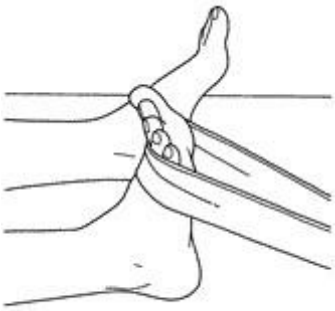
Post Ankle Fracture

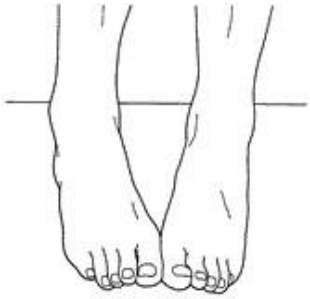
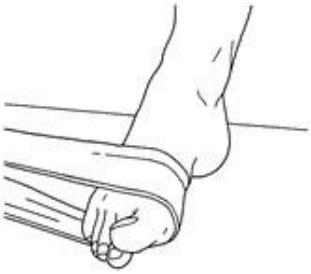
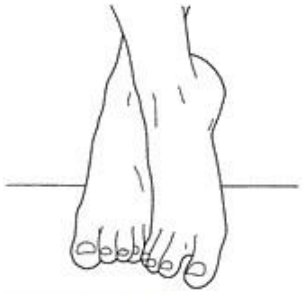
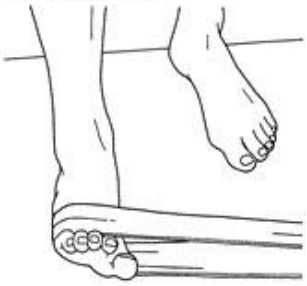
Early exercises – These are designed to help improve movement and function. They should be started as soon as your plaster is removed and carried out three to four times a day.

 ©PhysioTools Ltd	Lying on your back or sitting. Bend and straighten your ankles. If you keep your knees straight during the exercise you will stretch your calf muscles. Repeat 10 times
 ©PhysioTools Ltd	In sitting or lying. Move your ankle slowly in large circles. Repeat in opposite direction. Repeat 10 times.
 ©PhysioTools Ltd	Sitting on a chair. Alternate between raising your toes and raising your heels. Repeat 10 times.
 ©PhysioTools Ltd	Sitting on a chair. Cross the ankle to be stretched over the other knee. Place your hand on top of your foot and help to point your toes. This will stretch your ankle. You should feel the stretch in the front of your shin. Hold for 10 seconds. Repeat 10 times.

	When standing, place your foot on a chair. Line your heel up with the front edge of the chair. Hold the back of the chair for balance. Gradually move your knee towards the back of the chair keeping your whole foot in contact with the chair. This will stretch your ankle. You may feel a stretch in your calf and at the front of your ankle. Hold for 10 seconds. Repeat 10 times.
	Sit on the floor or on a chair with one leg out straight in front of you. Put a rubber exercise band or towel around your foot. Use the band / towel to gently pull your foot up towards your body. You will feel a stretch in your calf. Hold for 10 seconds. Repeat 10 times.
 	Sitting with your foot on the floor. Alternate between raising the inner border of your foot (big toe) and then the outer border (little toe). Hold for 10 seconds. Repeat 10 times. **Or**, sit on the floor or on a chair with one leg out straight in front of you. Put a rubber exercise band or towel around your foot. Use the band / towel to gently turn your foot in and out. Hold for 10 seconds. Repeat 10 times.

Intermediate exercises – Only start these exercises if your consultant is happy for you to put full weight on your foot.

 ©PhysioTools Ltd  ©PhysioTools Ltd	Sit on a chair or on the floor. Put the injured foot on top of the other foot. Try to point the toes of the foot that is on top while preventing any movement with the foot that is underneath. Hold for 10 seconds. Repeat 10 times. **Or**, sit on the floor or on a chair with one leg out straight in front of you. Tie a rubber exercise band to something secure and put the rubber exercise band around your injured foot. (Make sure there is some tension on the band to pull against). Pull your foot up towards your body against the resistance of the band. Hold for 10 seconds. Repeat 10 times.
 ©PhysioTools Ltd  ©PhysioTools Ltd	Sit on a chair or on the floor. Put the non-injured foot on top of the other foot. Try to lift the injured foot that is underneath. Hold for 10 seconds. Repeat 10 times. **Or**, sit on the floor or on a chair with one leg out straight in front of you. Put a rubber exercise band or towel around your foot. Pull the band / towel towards you to provide some resistance. Point your toes towards the floor, against the resistance of the band/ towel. Slowly return to starting position. Hold for 10 seconds. Repeat 10 times.

 ©PhysioTools Ltd  ©PhysioTools Ltd	Sit on a chair or on the floor. Put the inner borders of your big toes together. Press the inner borders of your big toes together. Hold for 10 seconds. Repeat 10 times. **Or**, sit on the floor or on a chair. Put a rubber exercise band around your foot. Turn your foot inwards against the resistance of the band as if to look at the sole of your foot. Hold for 10 seconds. Repeat 10 times.
 ©PhysioTools Ltd  ©PhysioTools Ltd	Sit on a chair or on the floor. Cross your feet and put the outer edges of your little toes together. Press the outer edges of your little toes together. Hold for 10 seconds. Repeat 10 times. **Or**, sit on the floor or on a chair. Put a rubber exercise band around your foot. Turn the foot outwards against the resistance band, as if to look at the outside of your foot. Hold for 10 seconds. Repeat 10 times.

	Practice standing on one leg. You may initially need to have fingertip support on the back of a chair or on a work surface. As your balance improves, repeat this without fingertip support. Hold for 10 seconds. Repeat 10 times.

References:

Royal Berkshire NHS Foundation Trust guidelines (retrieved 03/11/2022):

My Health Alberta:

Hip:
https://myhealth.alberta.ca/Health/aftercareinformation/pages/conditions.aspx?hwid=ug3861

Knee:

https://www.albertahipandknee.ca/assets/When-Youre-Having-a-Knee-Replacement.pdf


THR_Total_Hip_Replacement.pdf


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